TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 7 OF 10

# Take Away, Count Up




**Big idea:** The difference is a gap. To find how many are left, take away. To find how far apart two numbers are, count up the gap from the smaller number to the bigger one.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

## 1. At a glance

| | |
|---|---|
| **Strand** | Addition and Subtraction (Subtraction within 100 (tens and ones, no renaming); Subtraction as a gap: how many more, how many fewer; Relation between addition and subtraction (check by adding))<br>Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as MOE-aligned. Counting up as a subtraction method and the word difference sit inside the Primary 1 subtraction and inverse-relationship outcomes (docs/moe/syllabus_by_level.md, p. 31). The written renaming (borrowing) layout is not taught here. |
| **Objectives** | • Take away tens and then ones from a number within 100 (no renaming), and say what is left.<br>• Check a subtraction by adding the answer back (57 - 24 = 33 because 33 + 24 = 57).<br>• Find how many more one number is than another by counting up: to the next ten, by tens, then to the number.<br>• Choose a method: count up when the two numbers are close together, take away when the number taken away is small or a whole number of tens. |
| **Joining cold** | **Remember warm-up covers:** Bars and squares: a bar is a ten and a square is a one (tens and ones of a number); Fingers to make 10: how many more to reach the next ten; Count on in tens from any number (34, 44, 54).<br>Only counting to 20 and showing a number with fingers. The warm-up teaches the bars and squares again, so a child who missed Lessons 1, 5 and 6 can still join. Taking away small numbers within 10 is done with fingers or objects.<br>Lesson 7 is one of the hardest to miss in Term 1 (plan, section 2). A child who misses it watches the recording and does workbook pages 1 and 2 before Lesson 8. The warm-up carries every idea the lesson needs. |
| **Child materials** | • The printed workbook (pages 1 to 5) and the kit page (tens bars and ones squares), already cut out by a grown-up<br>• A pencil, a rubber and spare paper<br>• A pencil of a different colour for checking your own work<br>• An envelope or small cup to keep the bars and squares in |
| **Tutor materials** | • A slide deck and a visible clock or timer slide<br>• Your own set of tens bars and ones squares to show on camera if you wish<br>No ruler is needed. Nothing needs to be bought: the kit page is cut from the workbook. A grown-up does the cutting before the lesson; children never use scissors in the session. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears. |
| **Tiers** | Start here (7 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on. |
| **Contest-style move** | Used at Think harder only: difference as a gap (how many more); the gap stays the same when both numbers move by the same amount. Never advertised to children or parents. |

**Before the lesson**

- Solve every workbook question yourself first, both by taking away and by counting up.
- Tell parents a week before: print workbook pages 1 to 6, cut out the kit page (page 6) and keep the bars and squares in an envelope.
- Have your own bars and squares ready to show on camera, and have the slides open with the timer slide visible on screen share.
- Check the recording settings exclude children's video.

**Safeguarding and recording (this lesson)**

- The kit pieces stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. A grown-up cuts out the kit page before the lesson; children do not use scissors in the session.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Sneak-peek and exit answers are shown on the child's own paper to the camera at the same moment, once. There is no private chat and no board held up in front of the group.
- Never read out a child's name with a right or wrong answer. Never show a child's page in the recording; record the tutor and the screen only, with children's video excluded.
- A child who ticks I am still stuck gets a check-in in the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby). No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

## 2. Timing plan (90 minutes exactly)

| Minutes | Part | Min | Slides | Script fills |
|---|---|---|---|---|
| 0–2 | **Opening** | 2 | 1: Take Away, Count Up | 1:46 |
| 2–8 | **Remember warm-up** | 6 | 3: Warm-up: bars and squares, Fingers to ten, Count on to the next ten | 5:11 |
| 8–12 | **Sneak-peek puzzles** | 4 | 1: Sneak-peek puzzles | 3:37 |
| 12–28 | **Explore** | 16 | 3: Explore: take away, Explore: how far to go?, What did we notice? | 12:42 |
| 28–38 | **Mini-lesson** | 10 | 7: Take away tens, then ones, Check by adding, How many more? The gap, Count up in jumps, Your turn, Check: your turn, Choose your tool | 7:35 |
| 38–42 | **Brain break** | 4 | 1: Hop to the ten, jump the tens | 3:28 |
| 42–70 | **Workbook** | 28 | 5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page | 26:15 |
| 70–76 | **Talk time** | 6 | 2: Talk and explain, Back to the sneak-peek | 5:09 |
| 76–82 | **Exit ticket** | 6 | 2: Exit ticket, Check the exit ticket | 4:17 |
| 82–85 | **Look-back** | 3 | 1: Look-back | 2:28 |
| 85–90 | **Flex time and goodbye** | 5 | 1: Well done, inquirers! | 0:36 |
| | **Total** | **90** | | **73:04** |

The script needs 73:04 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 86:44. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: optional items and extra quick questions first, then the second explore round, then any quick-fire in the mini-lesson, then the optional slide. Never skip the look-back or the camera and privacy moments.

## 3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

### 0:00 – 2:00 Opening

script 1:46 of 2:00

| | | |
|---|---|---|
| 0:00 | SLIDE | Show slide **Take Away, Count Up** (s-title) |
| 0:25 | REC | *If you are watching the recording: hello! Get a pencil, paper, your workbook and your bars and squares. Pause or rewind any time.* |
| 0:36 | SAY | Good morning, inquirers! Cameras on, pencil ready, bars and squares on the table. This lesson is recorded for friends who are away. We record me and the screen only. Your camera may be off. |
| 0:54 | SAY | Two towers of blocks stand side by side. One is taller. |
| 0:59 | ASK | Tell your screen or your grown-up: how could we find how much taller? [wait about 20 s for answers] |
| 1:26 | SAY | Today we learn two tools for that: taking away, and counting up. |
| 1:32 | DO | Greet each child by first name and check their camera shows the page and hands. [0:15] |
| | NOTE | Routine and the four inquirer rules were introduced in Lesson 1; do not repeat them. Say the recording notice in your own words: the platform records the tutor and the shared screen only, and a child may keep the camera off. |

### 2:00 – 8:00 Remember warm-up

script 5:11 of 6:00

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|---|---|---|
| 2:00 | SLIDE | Show slide **Warm-up: bars and squares** (s-rem1) |
| 2:25 | REC | *Recording: join in! Say the numbers out loud and show your fingers. Pause if you need longer.* |
| 2:34 | SAY | A bar is a ten. A square is a one. Look at the picture. |
| 2:40 | ASK | How many bars, how many squares, and what number is it? [wait about 15 s for answers] |
| 3:01 | SAY | Four tens and six ones. We say forty-six. A few more, using fingers or whispers. |
| 3:08 | ASK | Two tens and seven ones. What number? [wait about 12 s for answers] |
| 3:24 | ASK | Five tens and no ones? [wait about 12 s for answers] |
| 3:38 | ASK | Eight tens and three ones? [wait about 12 s for answers] |
| 3:53 | SLIDE | Show slide **Fingers to ten** (s-rem2) |
| 4:18 | SAY | Fingers to ten. I say a number. Show me how many more fingers make ten. |
| 4:26 | ASK | Seven. How many more to make ten? [wait about 12 s for answers] |
| 4:41 | ASK | Four? [wait about 12 s for answers] |
| 4:54 | ASK | Nine? [wait about 12 s for answers] |
| 5:06 | ASK | Two? [wait about 12 s for answers] |
| 5:18 | SLIDE | Show slide **Count on to the next ten** (s-rem3) |
| 5:44 | SAY | Now count on in tens, starting at thirty-four. Clap on every number. |
| 5:50 | CHILDREN | Count on in tens together: 34, 44, 54, 64, 74. Then say the next ten after a few other starting numbers, such as 6 and 29. [0:30] |
| 6:20 | SAY | Next, how far to the next ten? It is the same trick as the fingers. |
| 6:27 | ASK | From thirty-eight, how many more to reach forty? [wait about 12 s for answers] |
| 6:43 | ASK | From forty-seven to fifty? [wait about 12 s for answers] |

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|---|---|---|
| 6:57 | ASK | From fifty-six to sixty? [wait about 12 s for answers] |
| | IF STUCK | If a child hesitates on 38 to 40 or 47 to 50, show the squares on the slide or your own kit: 38 is 3 bars and 8 squares, and 2 more squares fill the row. Do not move on until most children have answered; counting up to the next ten is the heart of the lesson. |
| | NOTE | Answers: 46, 27, 50, 83, then 3, 6, 1, 8, then 2, 3, 4. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20; the bars and squares are explained on the first slide. |

## 8:00 – 12:00 Sneak-peek puzzles

script 3:37 of 4:00

| | | |
|---|---|---|
| 8:00 | SLIDE | Show slide **Sneak-peek puzzles** (s-sneak) |
| 8:25 | REC | *Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.* |
| 8:34 | SAY | Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked. |
| 8:47 | READ | Puzzle one. There are 15 stars. Cross out 4 of them with your pencil. How many stars are left? |
| 8:57 | CHILDREN | Children cross out 4 stars and write how many are left on their own paper. [0:45] |
| 9:42 | READ | Puzzle two. Ali has 58 cards. Ben has 62 cards. How many more cards does Ben have? Write the number, and how you found it. |
| 9:56 | CHILDREN | Children write or draw their answer. [1:10] |
| 11:06 | HOLD UP | Everyone holds their paper to the camera at the same moment, once, after puzzle two. Do not read answers aloud. No private chat. |
| 11:31 | SAY | Thank you. We will come back to puzzle two a little later. |
| | NOTE | Puzzle 1 answer: 11 stars. Puzzle 2 answer: 4 more (count up 58 to 60 to 62, or 62 take away 58). Look for good ideas to praise later: a child who counted on, who drew two piles, or who used the next ten. Do not write scores or keep a list. |

## 12:00 – 28:00 Explore

script 12:42 of 16:00

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|---|---|---|
| 12:00 | SLIDE | Show slide **Explore: take away** (s-explore1) |
| 12:25 | REC | *Recording: pause now. Build each number with your bars and squares, then take away or add on. Press play when you have tried.* |
| 12:36 | SAY | Today your bars and squares are the teachers. A bar is a ten and a square is a one. First we take away. Build the number I say. |
| 12:50 | SAY | Build fifty-seven: 5 bars and 7 squares. |
| 12:54 | CHILDREN | Children build 57 with their kit. [0:40] |
| 13:34 | SAY | Take away 2 bars. Put them to one side. |
| 13:38 | CHILDREN | Children move 2 bars away. [0:20] |
| 13:58 | ASK | What number is left on the table? [wait about 25 s for answers] |
| 14:27 | SAY | Now take away 4 squares. |
| 14:30 | CHILDREN | Children move 4 squares away. [0:20] |
| 14:50 | ASK | What number is left now? [wait about 25 s for answers] |
| 15:17 | SAY | We took away 2 tens and then 4 ones. That is 24 altogether. Fifty-seven take away twenty-four leaves thirty-three. Tens first, then ones. |
| 15:28 | SAY | Your turn. Build seventy-six. Take away 2 bars and 5 squares. Write what is left on your paper. |
| 15:38 | CHILDREN | Children build 76, take away 2 bars and 5 squares, and write the number left. [1:20] |
| 16:58 | ASK | What is left? How did you take away? [wait about 40 s for answers] |

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|---|---|---|
| 17:42 | **SLIDE** | Show slide **Explore: how far to go?** (s-explore2) |
| 18:06 | **SAY** | New question. Ali has forty-seven stickers. Mei has fifty-two stickers. Build Ali's number. Then add pieces until you reach Mei's number. |
| 18:17 | **CHILDREN** | Children build 47, then add squares until they reach 52. [1:10] |
| 19:27 | **ASK** | How many pieces did you add? Tell me how you did it. [wait about 40 s for answers] |
| 20:13 | **SAY** | We added 3 squares to make a full row of fifty. Then 2 more squares to reach fifty-two. The gap from forty-seven to fifty-two is 5. |
| 20:26 | **SAY** | One more. Build thirty-eight. Add pieces until you reach sixty-two. Count what you add. |
| 20:33 | **CHILDREN** | Children build 38 and add squares and bars until they reach 62, counting what they add. [1:40] |
| 22:13 | **ASK** | What did you add? How many pieces is that altogether? [wait about 45 s for answers] |
| 23:03 | **SAY** | Two squares to reach forty. Two bars to reach sixty. Two squares to reach sixty-two. That is 2, and 20, and 2: twenty-four. |
| 23:14 | **SLIDE** | Show slide **What did we notice?** (s-explore3) |
| 23:40 | **ASK** | What did you notice? When did we take away, and when did we count up? [wait about 45 s for answers] |
| 24:32 | **SAY** | Taking away tells us what is left. Counting up tells us how big the gap is between two numbers. |
| | **IF STUCK** | A child who counts the pieces left instead of the pieces added: say, this time we are starting at the smaller number and adding on. Count the pieces you put down. |
| | **IF STUCK** | A child with too few squares: ten squares can be swapped for a bar, so there are always enough on the kit page. |
| | **NOTE** | Main room only: no breakout rooms, so you can see every camera. Pairs are optional: a grown-up may check the building. Take away: tens first, then ones. Gap: next ten, then tens, then ones. Answers: 37, 33, 51, 5, 24. |

## 28:00 – 38:00 Mini-lesson

script 7:35 of 10:00

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|---|---|---|
| 28:00 | **SLIDE** | Show slide **Take away tens, then ones** (s-m1) |
| 28:25 | **REC** | *Recording: keep watching and copy the examples. Rewind if you miss a step.* |
| 28:32 | **SAY** | Look. Fifty-seven is 5 tens and 7 ones. Take away 24. First the tens: fifty-seven take away twenty is thirty-seven. Then the ones: thirty-seven take away 4 is thirty-three. |
| 28:46 | **ASK** | When we took away tens, which digit changed, the tens or the ones? [wait about 15 s for answers] |
| 29:08 | **SLIDE** | Show slide **Check by adding** (s-m2) |
| 29:32 | **SAY** | Subtraction has a friend: addition. If thirty-three and twenty-four make fifty-seven, then we can check. Add the answer back. |
| 29:42 | **ASK** | If 33 + 24 makes 57, what is 57 take away 33? [wait about 15 s for answers] |
| 30:02 | **SLIDE** | Show slide **How many more? The gap** (s-m3) |
| 30:28 | **SAY** | Now a different question. Ali has 47 and Mei has 52. How many more does Mei have? The difference is the gap between the two numbers. Count up from 47: 3 to reach fifty, then 2 more. |
| 30:46 | **ASK** | What is the gap between 47 and 52? [wait about 15 s for answers] |
| 31:05 | **SLIDE** | Show slide **Count up in jumps** (s-m4) |
| 31:30 | **SAY** | A bigger gap: from 38 to 62. Hop to the next ten. Jump by tens. Hop to the number. Two, then twenty, then two. |
| 31:42 | **ASK** | How far is it from 38 to 62? [wait about 20 s for answers] |
| 32:06 | **SLIDE** | Show slide **Your turn** (s-m5) |
| 32:31 | **READ** | Your turn. Count up to find 81 take away 78. Take away to find 65 take away 23. Write both answers on your paper. |

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|---|---|---|
| 32:44 | **CHILDREN** | Children write two answers. [1:00] |
| 33:44 | **SLIDE** | Show slide **Check: your turn** (s-m5a) |
| 34:09 | **READ** | Three, and forty-two. 78 to 80 is 2, and 80 to 81 is 1. 65 take away 20 is 45, and 45 take away 3 is 42. |
| 34:24 | **SLIDE** | Show slide **Choose your tool** (s-m6) |
| 34:49 | **SAY** | Two tools. Count up when the two numbers are close together. Take away when you take away a small number or whole tens. Both tools give the same answer. |
| 35:03 | **ASK** | Which tool for 62 take away 59, and which for 62 take away 20? [wait about 25 s for answers] |
| | **IF STUCK** | If a child counts down by ones for 81 - 78, that works too: say, nice, and now let us see if counting up from 78 is shorter. |
| | **NOTE** | Key words every time: take away (what is left), count up (the gap), difference (the gap between two numbers). Do not teach the written borrowing layout. Do not teach exchanging a ten for ten ones: the Think harder page uses counting up instead. |

## 38:00 – 42:00 Brain break

script 3:28 of 4:00

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|---|---|---|
| 38:00 | **SLIDE** | Show slide **Hop to the ten, jump the tens** (s-break) |
| 38:25 | **REC** | *Recording: stand up and join in! Small hops to the next ten, big jumps for the tens.* |
| 38:34 | **SAY** | Stand up and get a sip of water. We hop to the next ten, then jump the tens. I say two numbers. You hop and jump the gap. |
| 38:48 | **CHILDREN** | Call 8 to 30, 25 to 40, 47 to 52, 36 to 71. Children hop the ones to the next ten, jump the tens, hop the last ones. Count the hops and jumps aloud together. [2:40] |

## 42:00 – 70:00 Workbook

script 26:15 of 28:00

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|---|---|---|
| 42:00 | **SLIDE** | Show slide **Workbook page 1: Start here** (s-ws1) |
| 42:25 | **REC** | *Recording: pause now and do workbook page 1. When you finish a page, press play to check it.* |
| 42:34 | **READ** | Page 1, Start here. Question 1: here is 68. Cross out 2 bars and fill in what is left. Question 2: take away. Question 3: take away the tens first, then the ones. |
| 42:52 | **CHILDREN** | Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:40] |
| 51:32 | **SLIDE** | Show slide **Check page 1** (s-ws1ans) |
| 51:57 | **READ** | Four tens and eight ones; forty-eight. Thirty-seven, fifty-three, fifty-six, eighty-one. Forty-eight, then forty-three. Mark your own page in another colour. |
| 52:08 | **SAY** | After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time. |
| 52:17 | **SLIDE** | Show slide **Workbook page 2: Start here** (s-ws2) |
| 52:42 | **READ** | Page 2. Question 4: take away, then check by adding. Question 5: Mei has 45 stickers and gives 13 away. Draw it. Question 6: Siti says 59 take away 20 is 57. Question 7: a bus has 64 seats and 21 have a child on them. |
| 53:07 | **CHILDREN** | Page 2, questions 4 to 7. [9:00] |
| 62:07 | **SLIDE** | Show slide **Check page 2** (s-ws2ans) |
| 62:32 | **READ** | Thirty-four and forty-six. Thirty-two. No: thirty-nine. Forty-three seats are empty. |
| 62:37 | **SLIDE** | Show slide **Choose your next page** (s-choice) |
| 63:02 | **SAY** | When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. |
| 63:15 | **CHILDREN** | Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [5:00] |

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|---|---|---|
| | IF STUCK | Q1: ask how many bars are left after you cross out 2. Q5: draw bars as tall rectangles and squares as small boxes. Q6: build 59 with the kit, take away 2 bars, and say what is left. Q7: ask, what does empty mean? Which tool will you use? |
| | IF STUCK | Think harder T1 to T4: ask, what is the next ten after the smaller number? How far is that? Then how many tens? Then the last hop. T3: draw two towers, one taller than the other, and add 10 to both. |
| | IF STUCK | A child who is still stuck ticks I am still stuck on the look-back page. You check in during the open session in the next lesson (a quick visit to the shared room or a hand signal, parent nearby). Never a private message. |
| | NOTE | Think harder is for count-up in a new setting. Go further asks which tool is quicker, and why. Do not announce who is on which page. Pairs are never split unless both children agree. |

## 70:00 – 76:00 Talk time

script 5:09 of 6:00

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|---|---|---|
| 70:00 | SLIDE | Show slide **Talk and explain** (s-talk) |
| 70:25 | REC | *Recording: pause. Tell a grown-up your answer to the talk question, and say why.* |
| 70:32 | READ | Ravi finds 52 take away 49 by taking away 40, then 9. Siti counts up from 49. They both get the same answer. Whose way is quicker? Why? |
| 70:47 | CHILDREN | Think for a moment. Use your bars and squares or your fingers if you like. [0:40] |
| 71:27 | CHILDREN | Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00] |
| 72:27 | ASK | Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers] |
| 74:03 | SLIDE | Show slide **Back to the sneak-peek** (s-sneakans) |
| 74:28 | SAY | Remember puzzle two? Ali has 58 cards and Ben has 62. The gap is 4: from fifty-eight to sixty is 2, and 2 more is sixty-two. |
| 74:41 | ASK | Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers] |
| | NOTE | Model answer: both answers are 3. Siti is quicker: 49 to 50 is 1, and 50 to 52 is 2. Ravi has to take away 40 and then 9 from 12, which is harder. When two numbers are close, counting up is short. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed. |

## 76:00 – 82:00 Exit ticket

script 4:17 of 6:00

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|---|---|---|
| 76:00 | SLIDE | Show slide **Exit ticket** (s-exit) |
| 76:25 | REC | *Recording: pause and do the exit ticket on your own. Then press play to check it.* |
| 76:33 | SAY | Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5. |
| 76:41 | CHILDREN | Children answer E1 to E3 on their own. [2:30] |
| 79:11 | HOLD UP | Everyone holds their ticket to the camera at the same moment so only you see it. No private chat. |
| 79:36 | SLIDE | Show slide **Check the exit ticket** (s-exitans) |
| 80:01 | READ | Thirty-four. Five, and five. Three: counting up is quicker because 87 and 90 are close. |
| 80:09 | SAY | Check your own ticket in a different colour. Well done. I will stamp your inquiry journal. |
| | NOTE | Mark S, D or H in the tutor tracker (tutor only): S = secure on all three, D = developing (one wrong), H = needs help (two or three wrong). If D or H, plan two minutes of catch-up at the start of Lesson 8, in the open session, no names. Do not read out any child's result. |

## 82:00 – 85:00 Look-back

script 2:28 of 3:00

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|---|---|---|
| 82:00 | SLIDE | Show slide **Look-back** (s-look) |
| 82:25 | REC | *Recording: pause and fill in your look-back box. You can pass.* |

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|---|---|---|
| 82:30 | **SAY** | Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. |
| 82:48 | **CHILDREN** | Children fill in the look-back box on page 5. [1:40] |

**85:00 – 90:00** **Flex time and goodbye** script 0:36 of 5:00

| | | |
|---|---|---|
| 85:00 | **SLIDE** | Show slide **Well done, inquirers!** (s-end) |
| 85:25 | **SAY** | Well done, inquirers. Next lesson is Balance and Swap. You need only a pencil and paper. Goodbye, and thank you to the grown-ups. |
| | **SLIDE** | Show slide **Go further: what if?** (s-opt) *(optional)* |
| | **SAY** | Optional, only if you are on time: the gap from 58 to 62 is 4. What is the gap from 59 to 63? And from 68 to 72? What do you notice? |
| | **CHILDREN** | Short think time, then take two answers. [1:30] |
| | **SLIDE** | Show slide **If we have time: pick one** (s-menu) *(optional)* |
| | **SAY** | Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each). |
| | **NOTE** | This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 4 and 4; when both numbers move up by the same amount, the gap stays the same. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on. |

## 3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask "How do you know?" and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

| Activity | Minutes | You say | Recording cue | Answer |
|---|---|---|---|---|
| **Is it always true?** | 4 (optional) | Mei says: if you add 10 to both numbers, the gap stays the same. Try 12 and 20, then 22 and 30, then 32 and 40. Is Mei always right? | *Recording: pause. Try the gap from 12 to 20, then 22 to 30. Is the gap the same?* | Yes. 12 to 20 is 8, 22 to 30 is 8, 32 to 40 is 8. Both numbers move up by the same amount, so the gap stays the same. |
| **Make a puzzle for your grown-up** | 4 (optional) | Write a short story about the gap between two numbers, such as two plants of different heights or two piles of cards. Give it to your grown-up to solve by counting up. | *Recording: pause. Write a gap story with two numbers for a grown-up to solve.* | No single answer. Check that the story has two numbers and asks how many more, and that counting up and adding back agree. |
| **Teach it back** | 3 (optional) | Pretend your grown-up or a toy has never counted up. Teach them how to find the gap from 58 to 62. | *Recording: pause. Teach a grown-up how to count up from 58 to 62.* | 58 to 60 is 2, and 60 to 62 is 2, so the gap is 4. Hop to the next ten, then hop to the number. |

# 4. Questions, answers, hints and common mistakes

## Sneak-peek puzzles (minutes 8 to 12)

**Puzzle 1 (access):** There are 15 stars. Cross out 4 of them with your pencil. How many stars are left?

**Answer: 11**

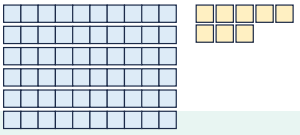
**Puzzle 2 (think):** Ali has 58 cards. Ben has 62 cards. How many more cards does Ben have? Write the number, and how you found it.

**Answer: 4 more. From 58 to 60 is 2, and from 60 to 62 is 2 more. 2 and 2 make 4.**

Unmarked. Papers are shown to the camera once, together. Reveal the answers where your script says (usually Talk time).

## Start here (workbook pages 1 and 2)

**S1** **Here is 68 as bars and squares. Cross out 2 bars. Fill in what is left.**



a) ___ tens and ___ ones are left
b) 68 - 20 = ___

**Answer: 4 tens and 8 ones are left; 68 - 20 = 48**

**Hint (say this first):** A bar is a ten. Cross out 2 bars, then count the bars and squares that are left.
**Worked solution:** 68 is 6 tens and 8 ones. Take away 2 tens: 4 tens and 8 ones are left. That is 48. The ones did not change.
**Common mistakes:** Crossing out 2 squares instead of 2 bars (66).; Writing 84 or 4 + 8 = 12.
Time for a middle child: about 2.5 min.

**S2** **Take away. Use your kit if you like.**

a) 57 - 20 = ___
b) 57 - 4 = ___
c) 86 - 30 = ___
d) 86 - 5 = ___

**Answer: 37, 53, 56, 81**

**Hint (say this first):** Taking away tens changes the tens digit. Taking away ones changes the ones digit.
**Worked solution:** (a) 5 tens take away 2 tens is 3 tens: 37. (b) 7 ones take away 4 ones is 3 ones: 53. (c) 8 tens take away 3 tens is 5 tens: 56. (d) 6 ones take away 5 is 1: 81.
**Common mistakes:** Changing the wrong digit (57 - 20 = 55).; Taking away from both digits.
Time for a middle child: about 3 min.

**S3** **Take away the tens first, then the ones. Find 68 - 25.**

a) 68 - 20 = ___
b) 48 - 5 = ___

**Answer: 48, then 43**

**Hint (say this first):** 25 is 2 tens and 5 ones. Take away the 2 tens first. Then take away the 5 ones from what is left.
**Worked solution:** 68 take away 20 is 48. Then 48 take away 5 is 43. So 68 - 25 = 43.
**Common mistakes:** Taking away 25 as 2 + 5 = 7 (61).; Forgetting to start part (b) from 48.
Time for a middle child: about 2.5 min.

**S4** **Take away. Then check by adding the answer back.**

a) 46 - 12 = ___
b) 34 + 12 = ___

**Answer: 34, then 46 (the same as the start, so it checks)**

**Hint (say this first):** Take away 1 ten, then 2 ones. To check, add 12 to your answer. Do you get 46 back?
**Worked solution:** 46 take away 10 is 36, and 36 take away 2 is 34. Check: 34 + 12 = 46, the number we started with. So 46 - 12 = 34 is right.
**Common mistakes:** Not adding back the number taken away (adding 46 + 12).; Writing 58 for the check.
Time for a middle child: about 3 min.

**S5** **Mei has 45 stickers. She gives 13 away. Draw 45 as bars and squares. Cross out 13. How many are left?**

Mei has ___ stickers left

**Answer: 32 (4 bars and 5 squares drawn; 1 bar and 3 squares crossed out)**

**Hint (say this first):** Draw 4 bars and 5 squares. Cross out 1 bar and 3 squares. Count what is left.
**Worked solution:** 45 is 4 tens and 5 ones. 13 is 1 ten and 3 ones. 45 take away 10 is 35. 35 take away 3 is 32. So Mei has 32 left.
**Common mistakes:** Crossing out 13 squares.; Writing 58 (adding instead of taking away).
Time for a middle child: about 3.5 min.

**S6** **Siti says 59 - 20 = 57. Is she right? Write the right answer, and say what she did.**

The right answer is ___

**Answer: No. 39. She took away 2 ones instead of 2 tens (57 is 59 - 2).**

**Hint (say this first):** Build 59 with your kit. Take away 2 bars, not 2 squares. What is left?
**Worked solution:** 59 is 5 tens and 9 ones. Take away 2 tens: 3 tens and 9 ones, which is 39. Siti took away 2 ones, so she got 57.
**Common mistakes:** Saying she is right.; Writing 79 (adding 20).
Time for a middle child: about 2.5 min.

**S7** **A bus has 64 seats. 21 seats have a child on them. How many seats are empty?**

___ seats are empty

**Answer: 43 seats**

**Hint (say this first):** Empty seats are the seats left when we take away the seats with a child. Take away the 2 tens first.
**Worked solution:** 64 take away 20 is 44. Then 44 take away 1 is 43. So 43 seats are empty.
**Common mistakes:** Adding 64 + 21 = 85.; Taking away only 2 or only 1.
Time for a middle child: about 3 min.

## Think harder (page 3): the taught idea in a new setting

**T1** **Mei has 62 beads. Ravi has 38 beads. How many more beads does Mei have? Count up from 38: to the next ten, then by tens, then to 62.**

a) 38 + ___ = 40
b) 40 + ___ = 60
c) 60 + ___ = 62
d) Mei has ___ more beads

**Answer: 2, 20, 2, so 24 more beads**

**Hint (say this first):** 38 needs 2 more to reach 40. Then jump by tens to 60. Then hop to 62. Add up the hops and the jump.
**Worked solution:** 38 to 40 is 2. 40 to 60 is 20. 60 to 62 is 2. The whole gap is 2 + 20 + 2 = 24. Check: 38 + 24 = 62.
**Common mistakes:** Forgetting that two tens is 20 (writing 2).; Writing 62 - 38 as 36 or 34.
Time for a middle child: about 4 min.

**T2** **Find 83 - 77 by counting up from 77.**

a) 77 + ___ = 80
b) 80 + ___ = 83
c) 83 - 77 = ___

**Answer: 3, 3, so 83 - 77 = 6**

**Hint (say this first):** What is the next ten after 77? How far is it? Then how far from that ten to 83?
**Worked solution:** 77 to 80 is 3. 80 to 83 is 3. The gap is 3 + 3 = 6. Check: 77 + 6 = 83.
**Common mistakes:** Writing only one hop (3).; Taking away 83 - 7 = 76.
Time for a middle child: about 3 min.

**T3** **Ali has 24 stickers. Ben has 31 stickers. Each child gets 10 more stickers. Now how many more stickers does Ben have than Ali?**

Ben has ___ more

> **Answer: 7 more (the gap stays the same: 34 and 41, and 41 - 34 = 7)**

**Hint (say this first):** Work out how many stickers each child has now. Or draw two towers and add the same 10 to both.
**Worked solution:** Ali now has 34 and Ben has 41. The gap from 34 to 41 is 7. It was 7 before too (24 to 31), because both children got the same 10 more.
**Common mistakes:** Writing 17 (7 + 10).; Writing 41 (Ben's number).
Time for a middle child: about 5 min.

**T4** **A book has 71 pages. Ravi has read 46 pages. How many pages are left to read? Count up from 46.**

___ pages are left

> **Answer: 25 pages (46 to 50 is 4, 50 to 70 is 20, 70 to 71 is 1)**

**Hint (say this first):** Start at 46. Hop to the next ten, jump by tens, then hop to 71. Add up all the hops and jumps.
**Worked solution:** 46 to 50 is 4. 50 to 70 is 20. 70 to 71 is 1. The gap is 4 + 20 + 1 = 25. Check: 46 + 25 = 71.
**Common mistakes:** Adding 71 + 46.; Forgetting the last hop of 1 (24).
Time for a middle child: about 4 min.

## Go further (page 4): mastery and extension

**G1** **Ravi wants to find 71 - 68 and 71 - 20. Which tool is quicker for each, take away or count up? Write why. Then find both answers.**

a) 71 - 68 = ___
b) 71 - 20 = ___

> **Answer: 71 - 68 = 3 (count up: the numbers are close). 71 - 20 = 51 (take away: 2 tens).**

**Hint (say this first):** For 71 - 68, how far is it from 68 up to 71? For 71 - 20, what happens to the tens digit?
**Worked solution:** 71 - 68: count up. 68 to 70 is 2 and 70 to 71 is 1, so 3. The numbers are close, so the gap is short. 71 - 20: take away 2 tens to get 5 tens and 1 one, which is 51. Counting up 20 would be a long way.
**Common mistakes:** Taking away 68 from 71 one at a time.; Counting up from 20 to 71 (too long).
Time for a middle child: about 6 min.

**G2** **A crate holds 100 eggs. 64 eggs are sold. How many eggs are left? Find it two ways, take away and count up. Do the two answers match?**

___ eggs are left

> **Answer: 36 eggs. Take away: 100 - 60 = 40, then 40 - 4 = 36. Count up: 64 to 70 is 6, then 70 to 100 is 30, so 36. They match.**

**Hint (say this first):** Take away: take away the 6 tens, then the 4 ones. Count up: hop from 64 to 70, then jump by tens to 100.
**Worked solution:** Take away: 100 - 60 = 40, and 40 - 4 = 36. Count up: 64 to 70 is 6, and 70 to 100 is 30, so 6 + 30 = 36. Both ways give 36. Check: 64 + 36 = 100.
**Common mistakes:** Writing 164 (adding).; Writing 40 (forgetting to take away the 4 ones).
Time for a middle child: about 6 min.

**G3** **Mei says 83 - 47 and 86 - 50 have the same answer. Is she right? Find both answers and say why.**

a) 83 - 47 = ___
b) 86 - 50 = ___

> **Answer: Yes. Both are 36. Both numbers go up by 3, so the gap does not change.**

**Hint (say this first):** Count up from 47 to 83 for the first. Take away 5 tens for the second. Then look at how 83 and 47 changed.
**Worked solution:** 83 - 47: 47 to 50 is 3, 50 to 80 is 30, 80 to 83 is 3, so 36. 86 - 50 = 36. 83 became 86 and 47 became 50: both went up by 3, so the gap between them stayed 36.
**Common mistakes:** Finding only one answer.; Saying the answers differ because the numbers differ.
Time for a middle child: about 6 min.

## Investigation (page 5): When is counting up quicker?

**Easy first step:** Find 52 - 49 and 52 - 10. Try take away and count up for each. Which way was quicker each time?
**Then:** Make up three subtractions where counting up is quicker, and three where taking away is quicker. What do you notice about the two numbers?

> **Model answer: Counting up is quicker when the two numbers are close together, so the gap is short (52 - 49 is 3). Taking away is quicker when the number taken away is small or a whole number of tens (52 - 10 is 42). If the numbers are far apart and the small number is awkward, both ways take longer.**

**Look for:** A child who tries several pairs, says what they notice about how far apart the numbers are, and gives a reason. Accept any clear rule that matches examples. A child who also checks by adding is doing well.
**Common mistakes:** Choosing a tool without looking at how far apart the numbers are.; Saying one tool is always quicker (it depends on the numbers).

## Talk and explain (minutes 70 to 76)

**Ravi finds 52 - 49 by taking away 40, then 9. Siti counts up from 49. They both get the same answer. Whose way is quicker? Why?** Sentence starter: I think ... because ...

> **Model answer: Siti's way is quicker. 49 to 50 is 1, and 50 to 52 is 2, so the answer is 3. The numbers are close together, so the gap is short. Ravi's way gets 3 too, but he has to take away 9 from 12.**

**Look for:** A reason about the numbers being close, with the hops (1 and 2). Praise any clear reason; both ways are right. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

## Exit ticket (minutes 76 to 82): 3 questions

**E1** **Take away.**
58 - 24 = ___

> **Answer: 34**

**Worked solution:** 58 take away 20 is 38, and 38 take away 4 is 34.
**Common mistakes:** Writing 82 (adding).; Writing 36 (took away 2 and 2).

**E2** **Count up from 57 to 62.**
a) 57 + ___ = 62
b) 62 - 57 = ___

> **Answer: 5, then 5**

**Worked solution:** 57 to 60 is 3, and 60 to 62 is 2. That is 5. So 62 - 57 = 5.
**Common mistakes:** Writing only the first hop (3).; Writing 15 (62 - 47).

**E3** **To find 90 - 87, which is quicker: take away or count up? Write the answer.**
90 - 87 = ___

> **Answer: 3. Counting up is quicker: 87 and 90 are close together.**

**Worked solution:** 87 to 90 is 3, so 90 - 87 = 3. Counting up is quick because the numbers are close.
**Common mistakes:** Taking away 87 from 90 one at a time.; Writing 13.

## Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (taking away tens and ones; hopping to the next ten; counting up in jumps; choosing a tool; something else); *What helped me* (bars and squares; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my tutor will come and help)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

## 5. Home practice and online practice

| # | Tier | Question | Answer |
|---|---|---|---|
| H1 | Start here | A jar has 78 sweets. 20 are eaten. How many sweets are left? | 58 |
| H2 | Start here | Take away, then check by adding: 76 - 43 = ___ . Check: ___ + 43 = 76. | 33; 33 + 43 = 76 |
| H3 | Start here | Take away the tens first, then the ones: 89 - 36 = ___ | 53 |
| H4 | Think harder | Ben has 47 cards. Ali has 71 cards. Who has more, and how many more? Count up. | Ali has 24 more (47 to 50 is 3, 50 to 70 is 20, 70 to 71 is 1) |
| H5 | Go further | At home: open a book at two page numbers, such as 38 and 71. Find how many pages from one to the other two ways: by taking away and by counting up. Do the two ways match? | Any two pages: for example 38 and 71: 71 - 38 = 33, and counting up 38 to 40 is 2, 40 to 70 is 30, 70 to 71 is 1, so 33. Both ways match. |

Online practice (`out/l07/practice`, key `t1l07`): 13 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

## 6. Parent line for the term report

Your child is learning to see numbers as tens and ones, to notice patterns and to say why. This week: taking away tens and ones, and counting up to find the gap between two numbers.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

## 7. Slide list

| # | Time | Slide | On screen | Speaker note |
|---|---|---|---|---|
| 1 | 0–2 | **Take Away, Count Up**<br>s-title | TODAY WE WILL: / Take away tens and ones, and count up to find a gap. | Lesson 7 of Term 1, 90 minutes. Objective: take away tens and ones within 100, count up to find the gap between two numbers, and choose the quicker tool. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (you and the screen only) and that a child may keep the camera off. |
| 2 | 2–8 | **Warm-up: bars and squares**<br>s-rem1 | A bar is a ten. A square is a one. / [picture: bundles 4 tens 6 ones] / 2 tens 7 ones 5 tens 8 tens 3 ones | Say: a bar is a ten, a square is a one. Children say the number for 4 bars and 6 squares, then the quick-fire three. Answers: 46, 27, 50, 83. This re-teaches tens and ones so a child who has not met them can join. No writing, no marks. |
| 3 | 2–8 | **Fingers to ten**<br>s-rem2 | [picture: ten_frame] / I show 7. How many more fingers make 10? / 7 4 9 2 | Say each number; children show on their fingers how many more make ten. Answers: 7 needs 3, 4 needs 6, 9 needs 1, 2 needs 8. Hands up on camera is fine. No writing, no marks. This is the number-bond idea you will use to hop to the next ten. |
| 4 | 2–8 | **Count on to the next ten**<br>s-rem3 | Start at 34. Count on in tens. Clap each one. / How many more from 38 to reach 40? / 38 ? 40 | Count on in tens from 34 (34, 44, 54, 64, 74) with a clap on each number. Then ask the next-ten questions: 38 to 40 is 2, 47 to 50 is 3, 56 to 60 is 4. Use the squares and a full row if a child is unsure. This is the hop you will use all lesson. |
| 5 | 8–12 | **Sneak-peek puzzles**<br>s-sneak | Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / There are 15 stars. Cross out 4 of them. / How many stars are left? / [picture: scatter] / PUZZLE 2: think: / Ali has 58 cards. / Ben has 62 cards. / How many more cards does Ben have? / Write the number, and how you found it. | Puzzle 1 (15 stars, cross out 4) is an access question: any child can try it by crossing out and counting. Puzzle 2 is a thinking question on the lesson idea, the gap between two numbers. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: 11; 4 more. Do not reveal Puzzle 2 until Talk time. Do not write scores. |
| 6 | 12–28 | **Explore: take away**<br>s-explore1 | Build 57: 5 bars and 7 squares. / Take away 2 bars. What is left? / Take away 4 squares. What is left? / Say: 57 take away 24 leaves ___. / YOUR TURN: / Build 76. Take away 2 bars and 5 squares. | Round 1, about 6 minutes. Objects first, no writing yet except the final number. Children build 57, take away 2 bars (37), then 4 squares (33). Then they build 76 and take away 2 bars and 5 squares (51). Ask how they took away. Key idea: tens first, then ones. A child with few squares can swap ten squares for a bar; the kit page has enough. |
| 7 | 12–28 | **Explore: how far to go?**<br>s-explore2 | Build the smaller number. Add pieces until you reach the bigger number. / [picture: bundles 4 tens 7 ones] / [picture: bundles 5 tens 2 ones] / 38 62 | Round 2, about 7 minutes, main room. Ali has 47 and Mei has 52: children add 3 squares to make 50, then 2 more to make 52, so the gap is 5. Then 38 to 62: 2 squares to make 40, 2 bars to make 60, 2 squares to make 62: 2 + 20 + 2 = 24. Children count what they add, not what they have. Ask them to say it as hop, jump, hop. |
| 8 | 12–28 | **What did we notice?**<br>s-explore3 | When did we take away? / When did we count up? / What does the gap tell us? | Take one or two answers from volunteers. Key idea: taking away tells us what is left; counting up tells us how far apart two numbers are, the difference. Repeat a child's answer in your own words so the recording picks it up. Both tools use the same bars and squares. |

| | | | | |
|---|---|---|---|---|
| 9 | 28–38 | **Take away tens, then ones**<br>s-m1 | [picture: bundles 5 tens 7 ones] / [picture: bundles 3 tens 3 ones] / 57 - 20 = 37. Then 37 - 4 = 33. | Think aloud: 57 is 5 tens and 7 ones. Take away 2 tens first: 37. Then take away 4 ones: 33. The picture on the right is what is left. Ask: which digit changes when we take away tens? Answer: the tens digit. Keep it to about a minute; children have just done this with the kit. |
| 10 | 28–38 | **Check by adding**<br>s-m2 | Take away \| Add to check / 57 - 24 = 33 \| 33 + 24 = 57 / Subtraction and addition are friends. | Show that the answer plus the number taken away gives back the start. 33 + 24 = 57, so 57 - 24 = 33 is right, and also 57 - 33 = 24. Ask: if 33 + 24 makes 57, what is 57 take away 33? Answer: 24. This is the relation between addition and subtraction, with no new vocabulary. |
| 11 | 28–38 | **How many more? The gap**<br>s-m3 | [picture: bundles 4 tens 7 ones] / [picture: bundles 5 tens 2 ones] / 47 +3 50 +2 52 / The gap is 3 + 2 = 5 | The difference between two numbers is the gap. Start at the smaller number and count up to the bigger: 47 to 50 is 3, 50 to 52 is 2, so the gap is 5. Ask: what is the gap between 47 and 52? Answer: 5. Say it as how many more: Mei has 5 more than Ali. |
| 12 | 28–38 | **Count up in jumps**<br>s-m4 | 38 +2 40 +20 60 +2 62 / Hop to the next ten. Jump by tens. Hop to the number. / 2 + 20 + 2 = 24 | Think aloud: from 38 hop 2 to reach 40; jump 2 tens to reach 60 (that is 20); hop 2 to reach 62. Add the hops and the jump: 2 + 20 + 2 = 24. Ask: how far is it from 38 to 62? Answer: 24. This is the same count as the explore with the bars and squares. |
| 13 | 28–38 | **Your turn**<br>s-m5 | Write on your paper. Timer: 1 minute / Count up: 81 - 78 = ___ / Take away: 65 - 23 = ___ | Each child writes on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. The first is a short gap, so counting up is quick; the second is a small number of tens and ones, so taking away is quick. |
| 14 | 28–38 | **Check: your turn**<br>s-m5a | 81 - 78 = 3: 78 to 80 is 2, then 1 more. / 65 - 23 = 42: 65 - 20 = 45, then 45 - 3 = 42. | Answers: 3 and 42. Read each answer aloud so a child watching the recording can mark their own work. Never ask a child to show a wrong answer to the group. |
| 15 | 28–38 | **Choose your tool**<br>s-m6 | Tool \| It is quick when / Count up \| the two numbers are close together / Take away \| you take away a small number or whole tens / Both tools give the same answer. | Ask: which tool for 62 take away 59, and which for 62 take away 20? Answer: count up for the first (3), take away for the second (42). This is the rule of thumb the Go further page asks children to explain. Both tools give the same answer; choosing is about being quick and being right. |
| 16 | 38–42 | **Hop to the ten, jump the tens**<br>s-break | Small hops to the next ten. Big jumps for the tens. / 8 to 30 / 25 to 40 / 47 to 52 / 36 to 71 / Sip some water. | Four minutes. Stand up on camera. Call each pair of numbers and have children hop the ones to the next ten, jump the tens, then hop the last ones; count the hops and jumps aloud. Gaps: 22, 15, 5, 35. No equipment. Keep it gentle for small spaces. |
| 17 | 42–70 | **Workbook page 1: Start here**<br>s-ws1 | Questions 1 to 3. Timer: 8 minutes / questions S1, S2, S3 | Read questions 1 and 2 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. |
| 18 | 42–70 | **Check page 1**<br>s-ws1ans | answers S1, S2, S3 | Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group. |
| 19 | 42–70 | **Workbook page 2: Start here**<br>s-ws2 | Questions 4 to 7. Timer: 9 minutes / questions S4, S5, S6, S7 | Read question 6 aloud: it is a spot-the-mistake. Read question 7 aloud as well. Children may draw bars and squares to decide. Question 4 asks them to check by adding. |
| 20 | 42–70 | **Check page 2**<br>s-ws2ans | answers S4, S5, S6, S7 | Read each answer aloud so a child watching the recording can mark their own work. Question 6 answer: no, 39, because she took away 2 ones instead of 2 tens. Never ask a child to show a wrong answer to the group. |

| | | | | |
|---|---|---|---|---|
| 21 | 42–70 | **Choose your next page**<br>s-choice | Page \| What it is / Think harder (page 3) \| Count up in a new setting / Go further (page 4) \| Which tool is quicker? / Investigation (page 5) \| When is counting up quicker? / Check \| Finish and check your own work / Everyone may choose. You can swap at any time. | Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 4 minutes. |
| 22 | 70–76 | **Talk and explain**<br>s-talk | RAVI AND SITI: / Ravi finds 52 - 49 by taking away 40, then 9. / Siti counts up from 49. / Whose way is quicker? Why? / I think ... because ... | One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: both get 3. Siti is quicker, because 49 and 52 are close: 49 to 50 is 1, and 50 to 52 is 2. Repeat each child's answer in your own words for the recording. |
| 23 | 70–76 | **Back to the sneak-peek**<br>s-sneakans | 58 +2 60 +2 62 / Ben has 4 more cards than Ali. | Reveal puzzle 2 now. Say: many of you wrote a good reason. Do not read out anyone's earlier answer. The gap from 58 to 62 is 2 and 2, which is 4. Some children will have taken away 62 - 58 or counted on one by one: all are good. |
| 24 | 76–82 | **Exit ticket**<br>s-exit | On your own. No looking back. Timer: 3 minutes / questions E1, E2, E3 | Children answer on page 5. After the timer everyone holds the ticket to the camera at the same moment, so only the tutor sees it (no private chat). Stamp the inquiry journal. |
| 25 | 76–82 | **Check the exit ticket**<br>s-exitans | answers E1, E2, E3 | Read each answer aloud. Children check their own ticket in a different colour. |
| 26 | 82–85 | **Look-back**<br>s-look | [look-back box] | Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, check in during the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby); never a direct message. At the start of the next lesson you may say to the whole group: today you could try drawing it, using objects or checking your answer. |
| 27 | 85–90 | **Well done, inquirers!**<br>s-end | Show a grown-up one thing you found out today. / NEXT LESSON: / Balance and Swap / You need a pencil and paper. | Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (stamps, the look-back). Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide. |
| 28 | optional | **Go further: what if?**<br>s-opt | The gap from 58 to 62 is 4. / What is the gap from 59 to 63? And from 68 to 72? / 58 62 59 63 68 72 | Optional, only if you are on time. Answer: 4 and 4. Both numbers move up by the same amount, so the gap stays the same. Take two answers and move on. This is the same idea as Think harder question 3. |
| 29 | optional | **If we have time: pick one**<br>s-menu | Activity \| What we do / Is it always true? \| Add 10 to both numbers. Does the gap stay the same? / Make a puzzle \| Write a gap story for your grown-up / Teach it back \| Show how to count up from 58 to 62 / Optional. Everyone may join. | Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum. |

## 8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

| Chapter | Part | You say (recording voice) | A child watching later |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 0:00 | Opening | *If you are watching the recording: hello! Get a pencil, paper, your workbook and your bars and squares. Pause or rewind any time.* | Gather your workbook, pencil, paper, and your bars and squares. Press play when ready. |
| 2:00 | Remember warm-up | *Recording: join in! Say the numbers out loud and show your fingers. Pause if you need longer.* | Play along. Say and show each answer before the video does. |
| 8:00 | Sneak-peek puzzles | *Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.* | Pause. Try both puzzles on your own paper. A guess is fine. |
| 12:00 | Explore | *Recording: pause now. Build each number with your bars and squares, then take away or add on. Press play when you have tried.* | Pause. Build, take away and add on with your bars and squares. Write what you find. |
| 28:00 | Mini-lesson | *Recording: keep watching and copy the examples. Rewind if you miss a step.* | Watch and copy the examples. Rewind any step you missed. |
| 38:00 | Brain break | *Recording: stand up and join in! Small hops to the next ten, big jumps for the tens.* | Stand up. Hop the ones and jump the tens. |
| 42:00 | Workbook | *Recording: pause now and do workbook page 1. When you finish a page, press play to check it.* | Pause. Do page 1, then play to check. Then page 2. Then choose a page to try. |
| 70:00 | Talk time | *Recording: pause. Tell a grown-up your answer to the talk question, and say why.* | Pause. Explain your answer to someone at home, with a reason. |
| 76:00 | Exit ticket | *Recording: pause and do the exit ticket on your own. Then press play to check it.* | Pause. Do the exit ticket alone, then play to check it. |
| 82:00 | Look-back | *Recording: pause and fill in your look-back box. You can pass.* | Pause. Fill in the look-back box, or pass. |
| 85:00 | Flex time and goodbye | *Recording: well done for finishing! Show a grown-up one thing you found out.* | Show a grown-up your workbook and tell them one thing you found out. |

## 9. Timing check (model, not a stopwatch)

| Part | Allocated | Script | Slow class | Flag |
|---|---|---|---|---|
| opening | 2:00 | 1:46 | 2:04 | ok |
| remember | 6:00 | 5:11 | 6:09 | ok |
| sneak | 4:00 | 3:37 | 4:15 | ok |
| explore | 16:00 | 12:42 | 15:03 | ok |
| mini | 10:00 | 7:35 | 8:56 | ok |
| break | 4:00 | 3:28 | 4:07 | ok |
| work | 28:00 | 26:15 | 31:21 | ok |
| talk | 6:00 | 5:09 | 6:07 | ok |
| exit | 6:00 | 4:17 | 5:05 | ok |
| look | 3:00 | 2:28 | 2:55 | ok |
| flex | 5:00 | 0:36 | 0:43 | ok |
| **Total** | 90:00 | 73:04 | 86:44 | |

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 7 questions; a middle child needs about 8.0 min for page 1 and 12.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.